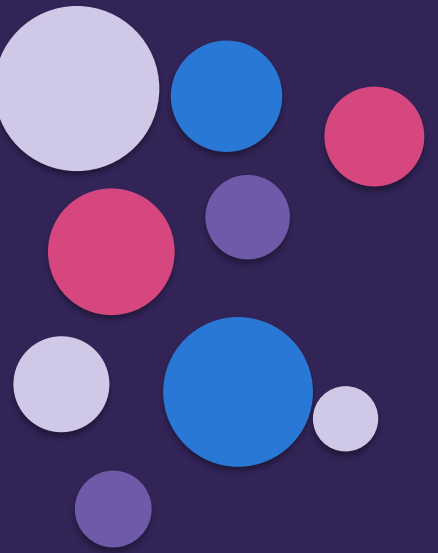
Nuclei Instance Segmentation on PanNuke



Progress report — Course Project 5

October 26, 2026 · bdd26 / jxh

The task

- Find every nucleus in H&E images
- Separate the ones that touch
- Classify each into 1 of 5 types

Dead (dying) nuclei: the hardest class

~7,900 image patches

19 tissue types

5 nucleus classes

How we score: PQ

higher = better · the official PanNuke metric

A prediction only counts when all three are right:

1

Found?
it is detected

2

Boundary right?
shapes overlap

3

Type right?
the class matches

Official 3-fold protocol **Last checkpoint** **No leaked weights**

Reproductions on PanNuke land at mPQ 0.46 – 0.50; every point is hard-won.

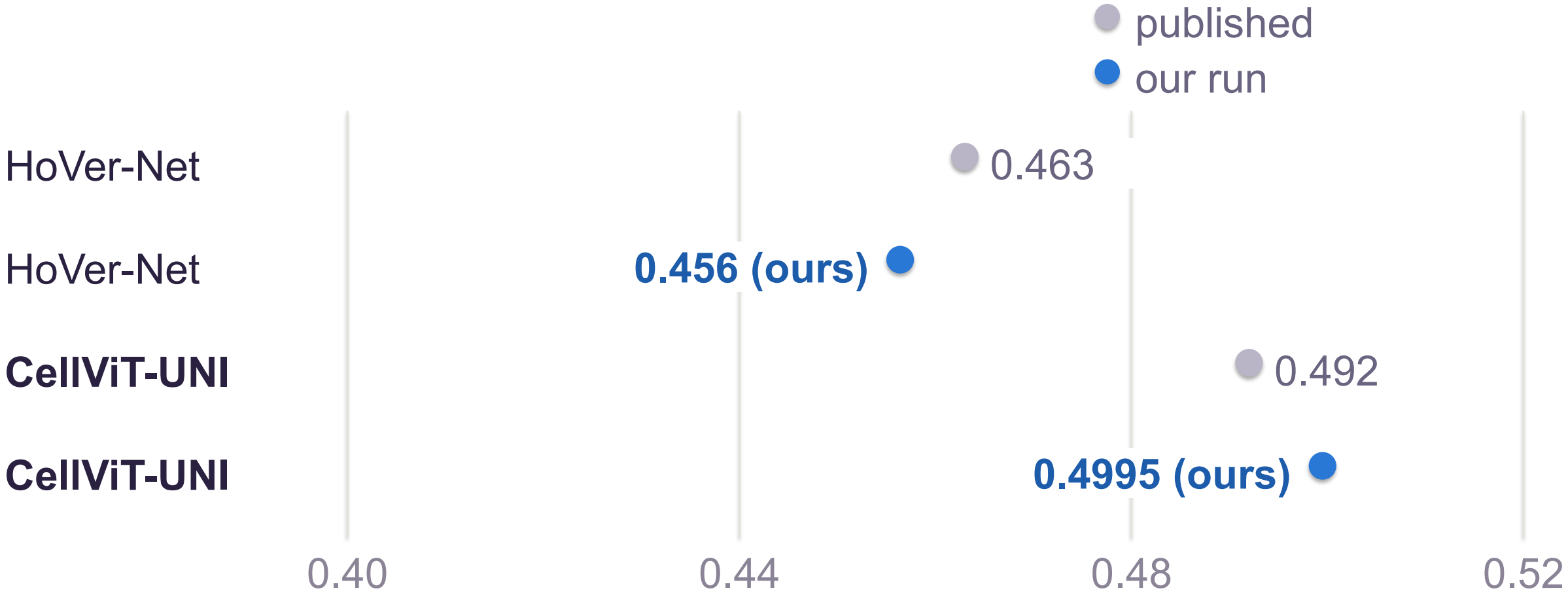
First: a pipeline you can trust

1. **Compact data pipeline —** streamed from zip: 37 GB → 3 GB
2. **Exact evaluator —** bit-identical to the official metric code
3. **One strict stack —** shared evaluator; provenance disclosed
4. **861 passed / 3 skipped (Oct 26 full CPU suite)**

Own training and released-weight protocols are reported separately.

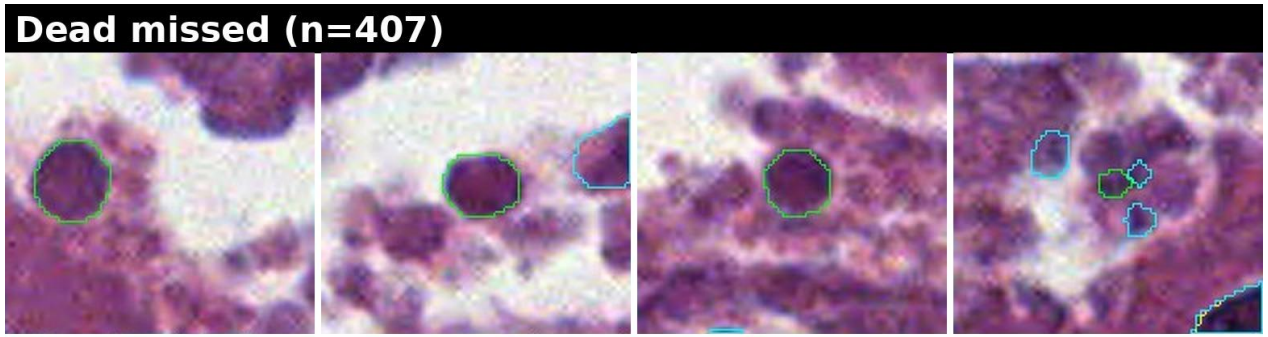
Baselines under the official protocol

mPQ, 3-fold mean, no TTA



Reference numbers are close; checkpoint provenance remains explicit.

A major baseline deficit: missed Dead nuclei



Green: ground-truth Dead nuclei the model missed entirely

38.5%

of Dead nuclei missed entirely (IoU < 0.1)

8 – 15%

missed rate, other classes

69% of matched Dead are typed correctly

→ **a detection-recall deficit**

Stage 1 — three new-paradigm attacks

Sep 25 – 28 · all judged against pre-registered rules

A

Language-anchored typing

CONCH VLM re-types nuclei from text

Null → diagnosis

B

Failure-driven synthesis

diffusion makes hard examples

Closed — no gain

C

Counterfactual stress test

re-render images, keep labels

Typing-only proxy

All three closed with informative negatives — and one key diagnosis.

Stage 1A — language anchoring (CONCH)

the model can name Dead nuclei — it just doesn't find them

0.695

zero-shot typing accuracy (CONCH pooling)

0.88

zero-shot Dead recall

- **Fused into the model:** test mPQ unchanged (0.4951 → 0.4953)
- **Also null:** loss re-weighting · balanced sampling · recovery

Tested re-typing recipes did not repair the detection deficit.

Stage 1B — generative synthesis (PixCell)

make more hard examples with a tissue generator

- **Mined the failures:** missed Dead are small & isolated; merges sit in same-class clusters
- **Copy-paste real Dead nuclei:** hurts — Dead PQ −0.031

- **In-context diffusion (1,146 images):** no gain — mPQ 0.4900, inside seed noise
- **Structural blocker:** mask-only conditioning cannot steer appearance by class

Verdict: closed

Honest scope: these recipes failed — other data interventions stay open.

Stage 1C — counterfactual stress testing

keep the labels, re-render the image → cheap robustness probe

- **Idea:** a generator as an evaluation simulator for stress testing
- **Result:** CF ranking vs real ranking is inverted (ρ = −0.89)
- **Mechanism:** typing rank agrees (ρ +1.00), detection gaps are not reproduced

Only three independent architectures; this renderer is not a transfer proxy.

Verdict: typing stress test, not a transfer proxy

Stage 2 — two attacks on the Dead deficit

Sep 28 – Oct 1 · detection-first architecture vs 2× resolution

Detection-first architecture

KongNet (CenterNet-style)

0.346 mPQ (−11 pts)

- touching same-class nuclei fuse
- interior-Dead miss: 0.50 vs 0.35

Negative

2× working resolution

512 px input & decoding

Dead ↑ at a priced cost

- interior-Dead miss 0.35 → 0.25
- cost: mPQ −0.009 · bPQ −0.007

Priced trade

Resolution helps Dead — can we keep the gain and drop the cost?

Stage 3 — reducing the bPQ trade-off

Oct 2 – 5 · four audited steps

1 Scale-aware decoding
fix decoder constants → recovers ~70% of the cost

2 M/S lever
val-selected merge + fragment fix — 9/9 runs

3 P2: gated additions
borrow 2× detections — Dead rises, guard fails

4 EF-P2: 30px filter
keep credible additions — near-zero bPQ tax

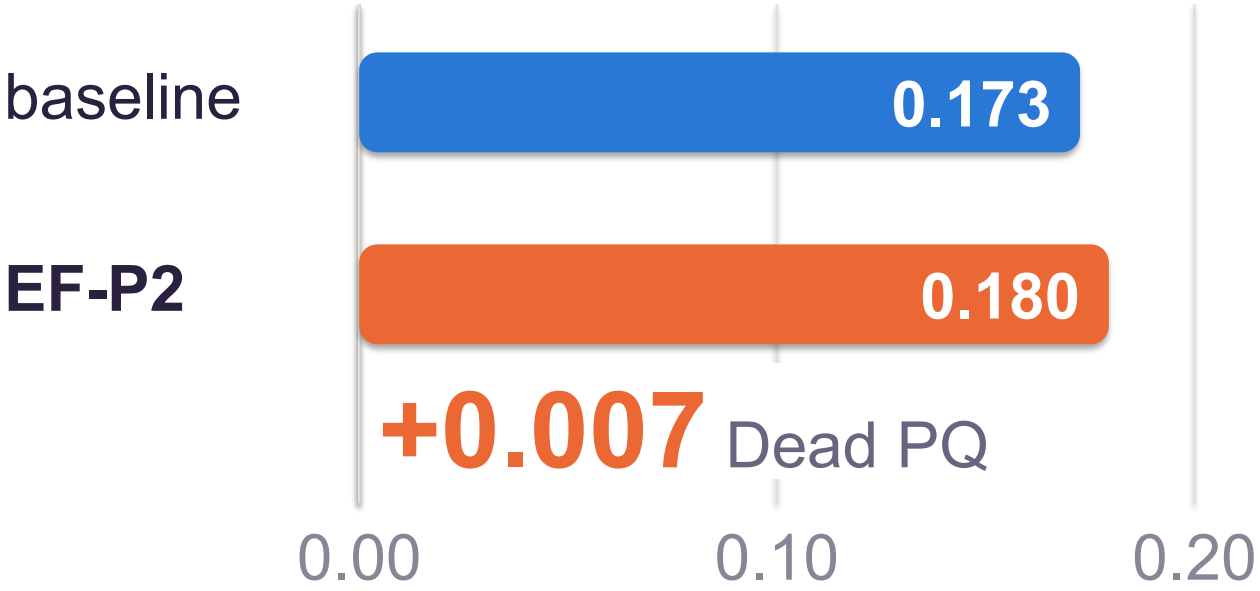
EF-P2 passes the pre-registered rule on 9 matched pairs.

Mean bPQ loss shrank to −.000009; this is not non-inferiority proof.

EF-P2 — a historical Dead-centred candidate

3-split mean over 9 matched (split, seed) pairs

Dead PQ — the target metric



0.4976 → 0.4983
mPQ (+0.0007)

0.6653 → 0.6653
bPQ (Δ ≈ 0 — the guard holds)

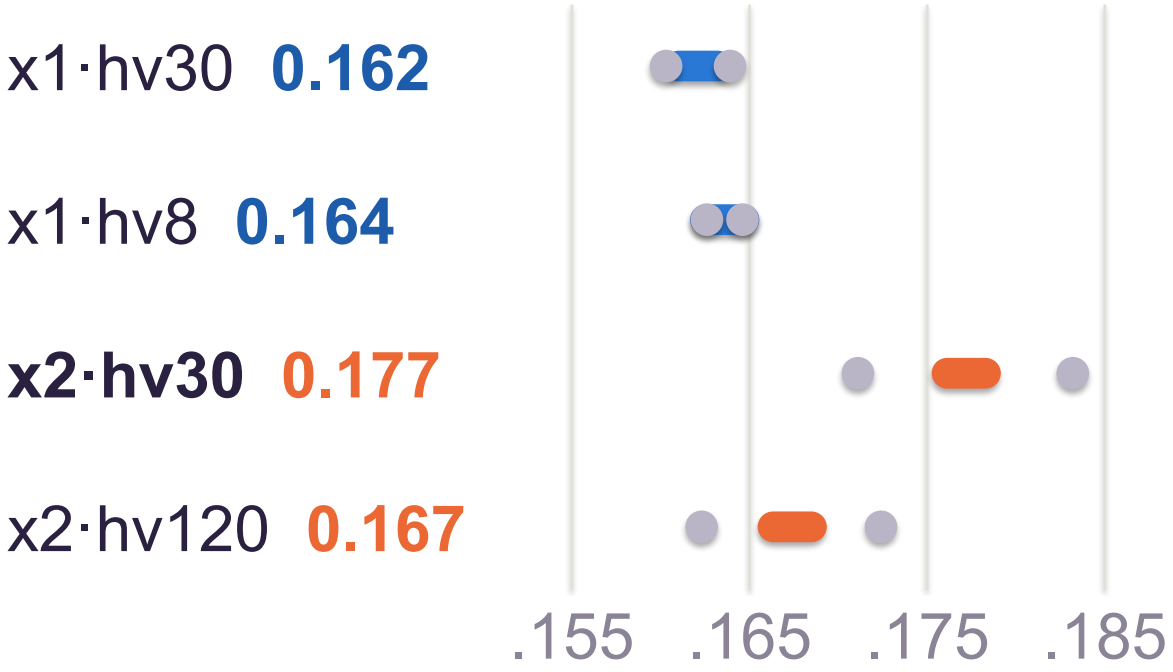
8 ↑ · 1 = · 0 ↓
per-pair Dead PQ signs

Candidate, not proof: CIs span zero · test folds reused across rounds

E2a: scale effects survive matched HV support

E2a · four arms × two seeds · 8/8 runs audited · validation fold only

Dead PQ by arm



gray dots: seeds · bar: arm mean

9.6%

of the 2× gain reproduced by the x1 cutoff change

4 / 4

matched-support pairs: strict Dead lower at 2×

HV threshold stays 30

8/8 runs hash-audited (audit − original = 0) · no test fold read

Dead specialist branch — closed before any training

DSB · four pre-registered gates · stop rule fired Oct 8

A **Base check** — base x1·hv30
gate-lowering explains 9.6% of the 2× gain

B **Conflict premise** — premise refuted
mean cosine +0.749 · 0/64 batches negative

C **Threshold decoupling** — too weak
peak +.0021 — under the +.005 close line

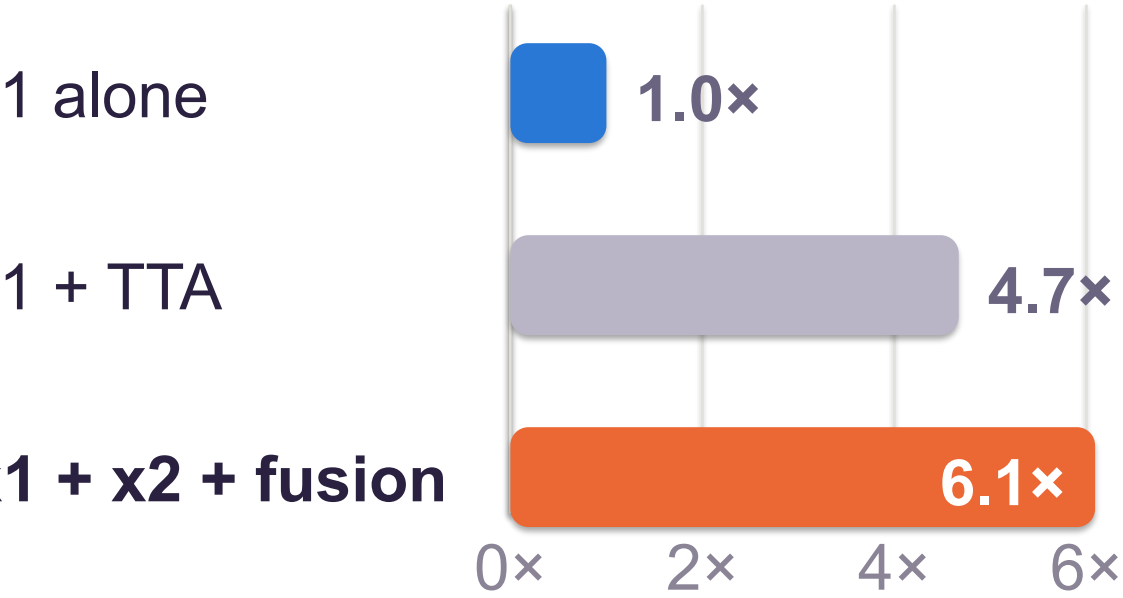
D **Oracle headroom** — headroom confirmed
+.156 Dead PQ if the 267 missed were found

Closed before expert training; gradient probe did use GPU compute.

What the two-model candidate costs

E0 · wall-clock on one shared A100 · unmodified inference path

Cost vs single-model x1



26 ms

per-image CPU fusion · frozen path replayed exactly

1.3×

over plain x1-TTA — which adds no Dead detections

batch 32 · shared GPU · absolute ms load-contaminated, ratios held 6.1–6.3

EF-P2 = mechanism evidence & teacher candidate — not a deployment win (pre-registered E0 rule).

Why trust these numbers

- **Pre-registered rules —** decision criteria fixed before any test data is read
- **Validation-only tuning —** last checkpoint; test folds reused across historical rounds
- **Paired seeds where specified —** seed noise measured: mPQ ±0.002, Dead PQ ±0.007
- **Honest negatives —** failed routes are reported as carefully as wins

Published ‘all-data’ checkpoints are never scored as test results.

Stage 5: typing headroom did not become a method

Oct 9–12 | fixed geometry, train/validation dev probe

GT-assisted oracle: mPQ .4995 → .6632
An upper bound with true labels, not an achievable method score.

T0–T3: frozen-feature read-outs closed
Best official mPQ change +.0002; Dead −.0053 trips the guardrail.

Interpretation
Simple post-hoc typing failed; this does not rule out all joint training.

Source: stage report V; old typing-oracle/T bootstrap CIs are not reused

Label quality: detector agreement, expert review absent

E1-C and E1-T | evidence on existence and typing

Existence: 66–73% vs 0.75% null
Zero-GT-overlap candidates with ≥3 cross-fold / cross-architecture supporters.

Typing: 72.5% base label vs 18.2% GT label
Jury votes on base mistypes; intrinsic flip rate 1.9%.

Evidence boundary
Correlated-detector resonance, not biological confirmation. Human review unavailable.

Source: stage report V / E1-C / E1-T; model pilot validated process only

Candidate utility and merged-label diagnostics

E3 closed at dev; M remains descriptive

E3 utility is predictable: AUC .721 / .737
Learned rows fail the frozen strict-Dead floor; identity fallback; line closed.

M: merging Inf/Conn changes official mPQ by +.0519
Largest response among 10 class pairs; oracle gap shrinks 35.2% in merged space.

Five-class method scores stay unchanged
A merged-label metric response is not recoverable performance or additive label-error loss.

Source: stage report V; RESEARCH_PLAN M entry; M used historical test artifacts

E4: corrected visibility probe passes its three gates

Corrected 2026-10-14 artifacts | single checkpoint, validation fold2 only

H1 area-standardized edge–interior Dead miss gap
+.1738 [.1060, .2528]

H2 difference-in-differences
+.1474 [.1392, .1557]; interpolation control −.0006

H3 fragment-area contrast
+.8175 [.7968, .8393]; does not prove HV zeroing causes missed nuclei

Source: E4 corrected results; old train-mode run and crude H1 gate superseded

Visibility supervision: four controlled dev arms

Seeds 19 / 1 | fold1 train, fold2 validation | 130 epochs, last checkpoint, x1, no TTA

A0: retrained baseline; A1: stronger crop exposure
RandomSizedCrop probability .1 → .5; HV min_size=30 in both.

A2: visible HV for newly truncated fragments
Same crops as A1; min_size=1 only for eligible truncated objects under 30 working pixels.

A3: min_size=1 for every visible object
Actual target difference is limited to objects below 30 working pixels.

Source: frozen visibility-supervision spec; A2−A1 isolates supervision

Dev validation: seed19 completed

Columns: official mPQ / bPQ / Dead PQ / strict Dead PQ | 2,523 images

A0 baseline **0.4840 / 0.6537 / 0.1611 / 0.1175**
A1 crop 0.4767 / 0.6465 / 0.1934 / 0.1254

A2 visible **0.4749 / 0.6474 / 0.1833 / 0.1199**
A3 low cutoff 0.4789 / 0.6469 / 0.1976 / 0.1183

A2 − A0 **-0.0091 / -0.0062 / +0.0222 / +0.0025**
Official Dead rises, while mPQ and bPQ both cross the frozen −.002 floor.

Source: per-run eval_fold2/summary.json | dev scores; no test or 3-split claim

Dev validation: the performance guardrail is violated

8/8 evaluations complete | original-image gate fails | 2026-10-15 06:06 EDT

Seed1 A0	**0.4784**	**0.6500**	**0.1575**	**0.1239**
Seed1 A1	0.4752	0.6450	0.2005	0.1337

Seed1 A2	**0.4750**	**0.6453**	**0.1844**	**0.1195**
Seed1 A3	0.4722	0.6439	0.1800	0.1178

Frozen dev admission cannot pass

Both seeds fail mPQ/bPQ floors; seed1 strict Dead falls .0044.

Snapshot only; no new runs, no menu expansion, no three-split confirmation

Intervention scope and interpretation

Label audit and logged training exposure are distinct measurements

A2 touches 7 Dead objects; A3 touches 70
One deterministic augmentation per fold1 image: 0.93% vs 9.3% of 751 Dead objects.

Actual training: 33,424 / 33,660 eligible exposures
All-class repeated events (seeds19/1); Dead-specific counts were not logged.

Integrate the final report
Separate official / strict scores, diagnostics and expert-review limits.

Source: fold1 audit and 130-epoch logs; exposures are not unique objects

C2: frozen-window mechanism endpoint

116 Dead / 38 images | fixed C2 windows | max-IoU < .1 misses

Seed19: A0 / A1 / A2 / A3 miss rates
0.474 / 0.440 / 0.466 / 0.457

Seed1: A0 / A1 / A2 / A3 miss rates
0.440 / 0.448 / 0.440 / 0.466

A1 minus A2: positive means fewer misses with selective HV
Mean -0.0086; 95% CI [-0.0409, +0.0177]

Paired tissue/image bootstrap on eligible images; 2 fixed seeds; no population claim

Mechanism evidence does not rescue the failed method

Original-image guards remain decisive | diagnostics are descriptive

Original-image interior Dead matched rate: A0 / A1 / A2
0.613 / 0.615 / 0.609

Original-image matched typing, all classes: A0 / A1 / A2
0.856 / 0.855 / 0.854

Decision: no added seeds or three-split confirmation
A2 fails original-image quality floors; C2 results cannot overturn that conjunction.

Full H1, split/merge, typing and paired-object records retained with input/code hashes

PM-Net feasibility: training-label geometry

Fold1 only | 2,656 original patches | no new model training

Query capacity: maximum 276 objects per patch
256 slots overflow by at least 20 objects in one Uterus patch; 512 covers this audit.

A rounded centroid is not always inside its mask
23 / 63,218 centers fall outside (1 / 967 Dead); 116 images contain zero objects.

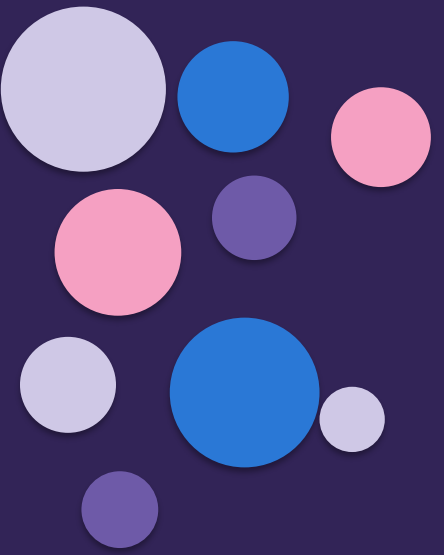
Before a new model: geometry and cost checks
Require legal inside-mask targets, empty-target handling, augmentation and cost audits.

Capacity audit is a lower bound, not a validated query budget or a performance result

October 15 checkpoint

- A strict, leak-free pipeline — baselines reproduced.
- Stages I–V complete; E4 supports a controlled dev test.
- EF-P2: +.007 Dead PQ at ~6.1× inference cost.
- **Visibility dev: A2 fails the performance guard.**

Subsequent query, point and proposal studies
are covered in the following update.



October 16-26 update follows

Stage 6: query masks did not become reliable instances

October 16-17 | fold1 training, fold2 development | two seeds, last checkpoint

E6 query masks: mPQ .1027 versus dense baseline .4814

The query model lost instance quality across all five classes and all 19 tissues.

E7 slower decay: mPQ .0880 versus baseline .4729

Changing decay .85 to .99 did not recover quality after 130 epochs.

Partial overlap is not a qualified instance

Localization, contour formation and separation remain distinct failure modes.

Source: stage report 6, E6/E7 full-training development studies

Query diagnostics: assisted fitting and predicted objects

E8-E18 | small-cohort mechanism studies

E8-E14: classifier, attention and matching repairs failed

Local queries, masked attention, auxiliary targets and exact matching did not pass the frozen gates.

E12 GT-assisted fitting: Dead 86.32% / 87.37%

True-object assistance demonstrates fitting capacity; it does not locate nuclei autonomously.

E17 emitted no proposals; E18 emitted too many

E18 final-instance audit precision was about 3.5-3.7%; point and instance metrics differ.

Source: stage report 6; E7-E18 closeout; individual diagnostic records

High-resolution features: the tested adapter failed

E19 | zero-initialized CNN residual at the finest CellViT decoder skip

Interior Dead matching change: -.0081

Two-seed mean; 95% CI [-.0226, +.0074]. The primary criterion failed.

Official Dead PQ -.0018; strict Dead PQ +.0081

Different metric responses cannot replace the selected primary endpoint.

Close this fixed single-skip adapter recipe

These results do not establish that every high-resolution architecture must fail.

Source: HIRES_ADAPTER_E19_DEV_2026-10-18; stage report 6

Point proposals: fitting and audit protocol

P3 onward | repeated fold1 development data | 130 epochs, last checkpoint

2,134 fitting images; 522 audit images

Audit support: 11 Dead-positive images and 145 internal Dead objects; 116 are Lung/Colon.

Unique geometric matching within 3 native pixels

These are class-agnostic candidate points evaluated on the internal Dead GT subset.

Original conjunction must pass in both seeds

Dead recall >= .60, gain >= .05, positive lower CI, precision >= .75, plus coverage guards.

Source: proposal P3 protocol and scripts/proposal_p3_utils.py

P3-P9: no recipe passed its original gate

Treatment minus its own fresh control | internal Dead point recall

Round / treatment	Seed 19	Seed 1	Admission
P3 adaptation + augmentation	+.068966	+.013793	failed
P4 foreground BCE	-.027586	+.006897	failed
P5 native-grid refinement	.000000	-.006897	failed
P6 wider Gaussian suppression	+.020690	+.020690	failed
P7 training RGB gains	+.034483	-.013793	failed
P8 image normalization	-.034483	-.048276	failed
P9 Dead-positive weight 4	+.006897	+.048276	failed

Source: PROPOSAL_PROGRESS_2026-10-26 and linked frozen closeouts

P10-P15: less adaptation did not ensure transfer

Treatment minus its own fresh control | internal Dead point recall

Round / treatment	Seed 19	Seed 1	Admission
P10 last 4 encoder blocks	+.027586	-.013793	failed
P11 positive-mass redistribution	+.041379	.000000	failed
P12 decoder dropout .2	-.034483	-.027586	failed
P13 rare-positive replacement	-.137931	-.048276	failed
P14 encoder LR 1e-6	-.068966	-.089655	failed
P15 integer translation +/-16	-.193103	-.089655	failed

Source: PROPOSAL_PROGRESS_2026-10-26 and linked frozen closeouts

Two failure profiles in the proposal studies

Chronology of hypotheses sharing audit data; not a hyperparameter ranking

P13: near-perfect training fit, weak audit recall

Fit internal Dead .9985 / 1.0000; audit .3517 / .3586. Extra exposure did not transfer.

P15: both fit and audit coverage fell

Audit precision rose to .865 / .863 while overall recall fell by .203 / .214.

Precision and recall must be read together

Fewer candidate points can raise precision without solving the missed-object problem.

Source: P13 and P15 final results; stage report 6

P16: Gaussian regression completed all four runs

Array 55287551 | all 130 epochs, COMPLETED / 0:0 | checked October 26

Seed / objective	Dead hits / 145	Overall recall	Precision
19 / original focal C	70 = .4828	.5980	.6941
19 / squared-error R	75 = .5172	.6025	.7011
1 / original focal C	63 = .4345	.5940	.6951
1 / squared-error R	72 = .4966	.6009	.7050

Source: recovered P16 per-image results, summaries and 130-epoch histories

P16: positive point estimates, failed admission

Original 1,000-draw tissue-stratified image bootstrap | same fixed cohort

Seed 19: +.0345, 95% CI [-0.0341, +0.1311]

Gain is below .05; CI spans zero; absolute recall and precision are below the gates.

Seed 1: +.0621, 95% CI [+0.0282, +0.1150]

Gain and CI conditions pass; absolute recall .4966 and precision .7050 still fail.

Decision: close the fixed Gaussian-regression recipe

Both seed-level conjunctions fail. Do not expand coefficient, sigma or threshold menus.

Source: original admission function and independent bootstrap replay, October 26

P17: convolutional detector versus adapted UNI

Independently frozen before P16 seed 1 finished | C/K x seeds 19/1

K: a three-stage CNN pyramid trained from scratch

Widths 32/64/128, multiscale fusion and a native-center heatmap; no queries or masks.

Keep the original cohort, decode and admission rule

Four fresh 130-epoch runs. Architecture, pretraining and learning rate change together.

V100 smoke passed; production result is pending

291 node checks and strict readiness passed. This is a recipe comparison, not pure locality attribution.

Source: P17 frozen specification and submission record; smoke 55297045

P17 live training status

Snapshot: October 26, 11:29 EDT | array 55297061 | concurrency 2

Task / run	State	Epochs / 130	Elapsed
0 / C seed 19	RUNNING	44	~1 h 26 min
1 / K seed 19	RUNNING	59	~1 h 12 min
2 / C seed 1	PENDING	not started	array limit
3 / K seed 1	PENDING	not started	array limit

Source: existing Expanse session; squeue/sacct and run status.json; dated snapshot

What the evidence currently supports

October 26 update | stage reports I-VI and current experiment snapshot

A reproduced baseline and a documented Dead detection deficit

EF-P2 retains a historical +.007 Dead PQ signal at about 6.1x inference cost.

No P3-P16 proposal recipe earned expansion

P16 has completed and fails the original gate; P17 remains in progress.

Next: finish P17, verify provenance, apply the same rule

Reused development points are not mask PQ, test results or patient-independent evidence.

Sources: stage reports 3-6; original closeouts; P16 replay; P17 dated status